

HOMEWORK 2

AUTHOR

Shey Dorji, Alecia Dezzani

GETTING STARTED: ACCESS DATA & PACKAGES

```
library(tidyverse)
library(lubridate)
library(here)
library(ggplot2)
library(jtools)
library(knitr)
library(kableExtra)
```

systemfonts and textshaping have been compiled with different versions of Freetype. Because of this, textshaping will not use the font cache provided by systemfonts

```
library(janitor)
library(geomtextpath)
```

```
#install.packages("dataRetrieval")
#library(dataRetrieval)

#commenting out loading the data from NWIS
#siteNumbers <- c("14211720", "14211010", "14166000")
#parameterCd <- "00010" # water temperature
#startDate <- "2010-01-01"
#endDate <- "2023-12-31"

#temp_data <- readNWISdv(siteNumbers, parameterCd, startDate, endDate)
```

GENERATING DATA

```
# rename columns for clarity
#colnames(temp_data) <- c("agency_cd", "site_no", "Date", "temp_C")

# extract some date info for use later
#temp_data$month = month(temp_data$Date, label=TRUE)
#temp_data$year = year(temp_data$Date)

# check by plotting
#ggplot(temp_data, aes(x=Date, y=temp_C, color=site_no)) +
#  #geom_line() +
#  #labs(title="Stream Temperature at Willamette River Sites",
```

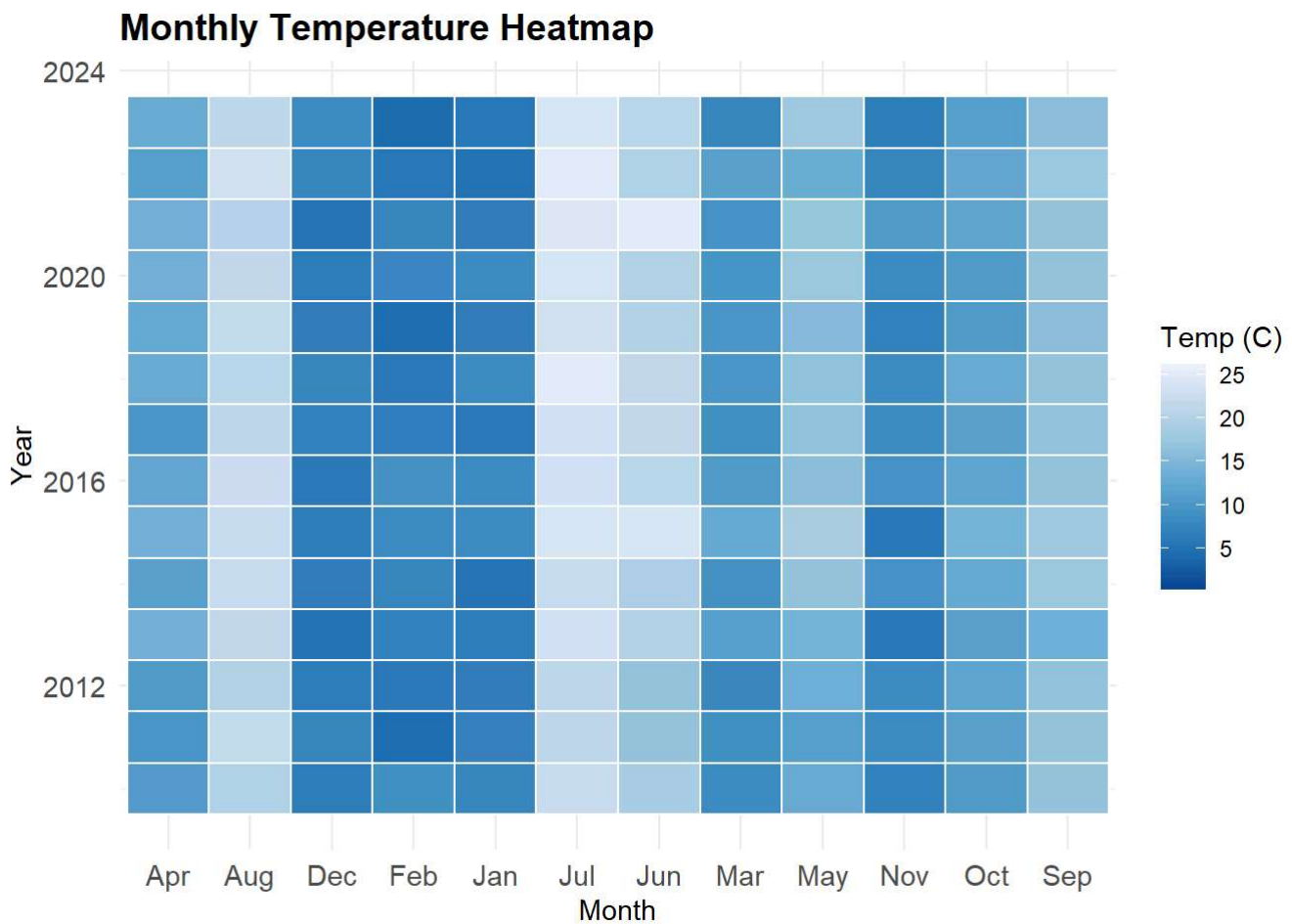
```
# x="Date", y="Temperature (°C)", color="Site Number") +  
#theme_minimal()
```

SAVE & LOAD DATA FRAME

```
#save the dataframe  
#write_csv(temp_data, here("temp_data.csv"))  
  
#local data load  
temp_data_final <- read_csv(here("temp_data_final.csv"))
```

PLOT WATER TEMPERATURE DATA

```
ggplot(temp_data_final, aes(x = month, y = year, fill = temp_C)) +  
  geom_tile(color = "white", linewidth = 0.5) +  
  scale_fill_distiller(palette = "Blues", name = "Temp (C)") +  
  labs(  
    title = "Monthly Temperature Heatmap",  
    x = "Month",  
    y = "Year",  
    fill = "Temperature (C)"  
  ) +  
  theme_minimal() +  
  theme(  
    axis.text = element_text(size = 11),  
    plot.title = element_text(face = "bold", size = 14)  
  )
```



PART I: ANALYSIS OF AUGUST WATER TEMPERATURES

A) IS THERE A TREND IN TEMPERATURE IN THE MONTH OF AUGUST?

```
# get data for August
aug_data <- temp_data_final |>
  clean_names() |>
  filter(month == "Aug")

# is there a trend between year and temperature? Is the temperature changing over time?
aug_lm <- lm(temp_c ~ year, data = aug_data)
summ(aug_lm, digits = 4)
```

Observations	1299
Dependent variable	temp_c

Type		OLS linear regression			
F(1,1297)		113.4532			
R²		0.0804			
Adj. R²		0.0797			
		EST.	S.E.	T VAL.	P
(Intercept)		-284.4079	28.6214	-9.9369	0.0000
year		0.1512	0.0142	10.6514	0.0000

Standard errors: OLS

Yes, there is a significant relationship between the year and the water temperature in August. The p-value of the coefficient `year` is 0.0000 which is less than our alpha threshold of 0.05.

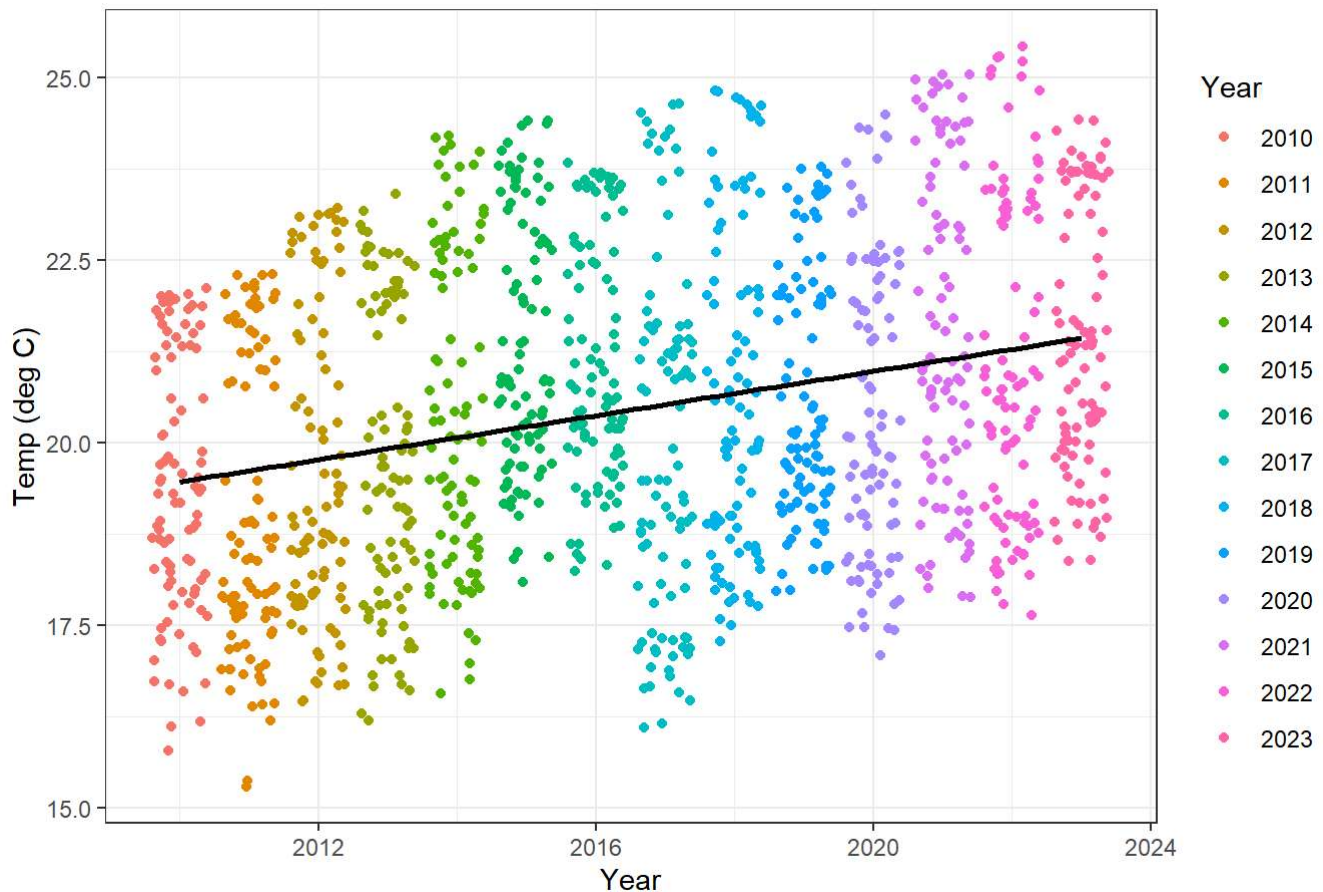
B) IS THE TREND POSITIVE (INCREASING) OR NEGATIVE (DECREASING)?

```
aug_plot <- ggplot(aug_data, aes(x = year, y = temp_c, color = as.factor(year))) +
  theme_bw() +
  geom_jitter() +
  geom_smooth(aes(group = 1),
              method = "lm",
              se = FALSE,
              color = "black") +
  labs(title = "August Recorded Water Temperature for all sites",
       x = "Year",
       y = "Temp (deg C)",
       color = "Year")

aug_plot
```

``geom_smooth()`` using formula = `'y ~ x'`

August Recorded Water Temperature for all sites



Based on the plot above and the regression coefficient = 0.1512, the linear regression estimates that the August water temperature is increasing by 0.1512 degrees Celsius every year. However, this linear model has an $R^2 = 0.0804$ and the figure shows clearly that the data is not obviously linear. Therefore, only 8% of the variation in temperature can be attributed to a difference in year, and year is not be the best variable to estimate or predict change in water temperature.

C) ARE THERE DIFFERENT TRENDS IN EACH SITE?

```
# OLS regression for Site 14211720
aug_1720 <- aug_data |>
  filter(site_no == 14211720)

aug_1720_lm <- lm(temp_c ~ year,
                  data = aug_1720)
summ(aug_1720_lm, digits = 4, model.fit = FALSE, model.info = FALSE)
```

	EST.	S.E.	T VAL.	P
(Intercept)	-271.4474	23.3695	-11.6155	0.0000
year	0.1459	0.0116	12.5926	0.0000

Standard errors: OLS

```
# OLS regression for Site 14211010
aug_1010 <- aug_data |>
  filter(site_no == 14211010)

aug_1010_lm <- lm(temp_c ~ year,
                  data = aug_1010)
summ(aug_1010_lm, digits = 4, model.fit = FALSE, model.info = FALSE)
```

	EST.	S.E.	T VAL.	P
(Intercept)	-335.0182	26.8284	-12.4875	0.0000
year	0.1760	0.0133	13.2299	0.0000

Standard errors: OLS

```
# OLS regression for Site 14166000
aug_6000 <- aug_data |>
  filter(site_no == 14166000)

aug_6000_lm <- lm(temp_c ~ year,
                  data = aug_6000)
summ(aug_6000_lm, digits = 4, model.fit = FALSE, model.info = FALSE)
```

	EST.	S.E.	T VAL.	P
(Intercept)	-243.0310	26.1096	-9.3081	0.0000
year	0.1297	0.0129	10.0208	0.0000

Standard errors: OLS

```

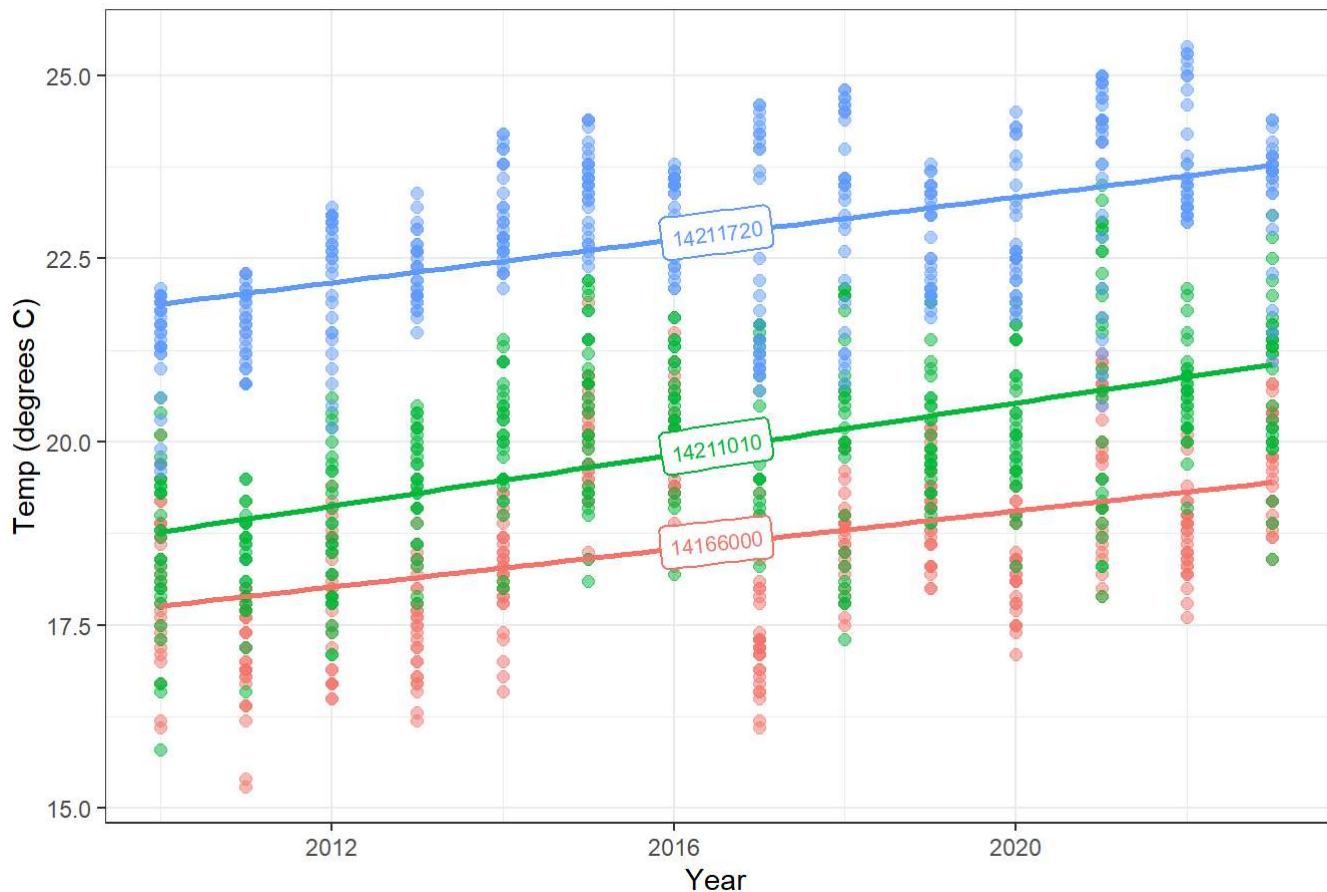
aug_site_plot <- ggplot(aug_data, aes(x = year,
                                     y = temp_c,
                                     color = as.factor(site_no))) +

  theme_bw() +
  geom_point(size = 2, alpha = 0.5) +
  geom_labelsmooth(aes(label = site_no),
                  fill = "white",
                  method = "lm",
                  formula = y ~ x,
                  size = 2.75,
                  linewidth = 1,
                  boxlinewidth = 0.4) +
  labs(title = "August Water Temperature Trends Per Site",
       x = "Year",
       y = "Temp (degrees C)",
       color = "Site") +
  guides(color = 'none')

aug_site_plot

```

August Water Temperature Trends Per Site



Yes, there are differences in temperature trends between sites. At site 14211720, the linear model estimates that the water increases by 0.1459 degrees C per year. At site 14211010, the linear model estimates that the water increases by 0.1760 degrees C per year. At site 14166000, the linear model estimates that the water increases by 0.1297 degrees C per year. The figure above (*"August Water Temperature Trends Per Site"*) shows that each site has its own temperature trends. These results make sense, as each site likely has its own unique combination of upstream water flow, water depth, sun exposure, flow speed, et cetera.

PART II: ANALYSIS OF DECEMBER WATER TEMPERATURES

```
#GLM for predicting stream temperature in December

temp_data_final$temp_C <- as.numeric(temp_data_final$temp_C)
temp_data_final$year <- as.numeric(temp_data_final$year)

# filter for dec temperatures
dec_data <- subset(temp_data_final, month == "Dec")

# fit a GLM model to see if there is a trend in dec temperatures over time, separated
```



```
dec_model <- glm(temp_C ~ year*site_no, data = dec_data)
summary(dec_model)
```

Call:

```
glm(formula = temp_C ~ year * site_no, data = dec_data)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.615e+04	1.382e+04	1.168	0.243
year	-7.889e+00	6.855e+00	-1.151	0.250
site_no	1.145e-02	0.727e-01	1.176	0.240
year:site_no	5.596e-07	4.829e-07	1.159	0.247

(Dispersion parameter for gaussian family taken to be 2.24653)

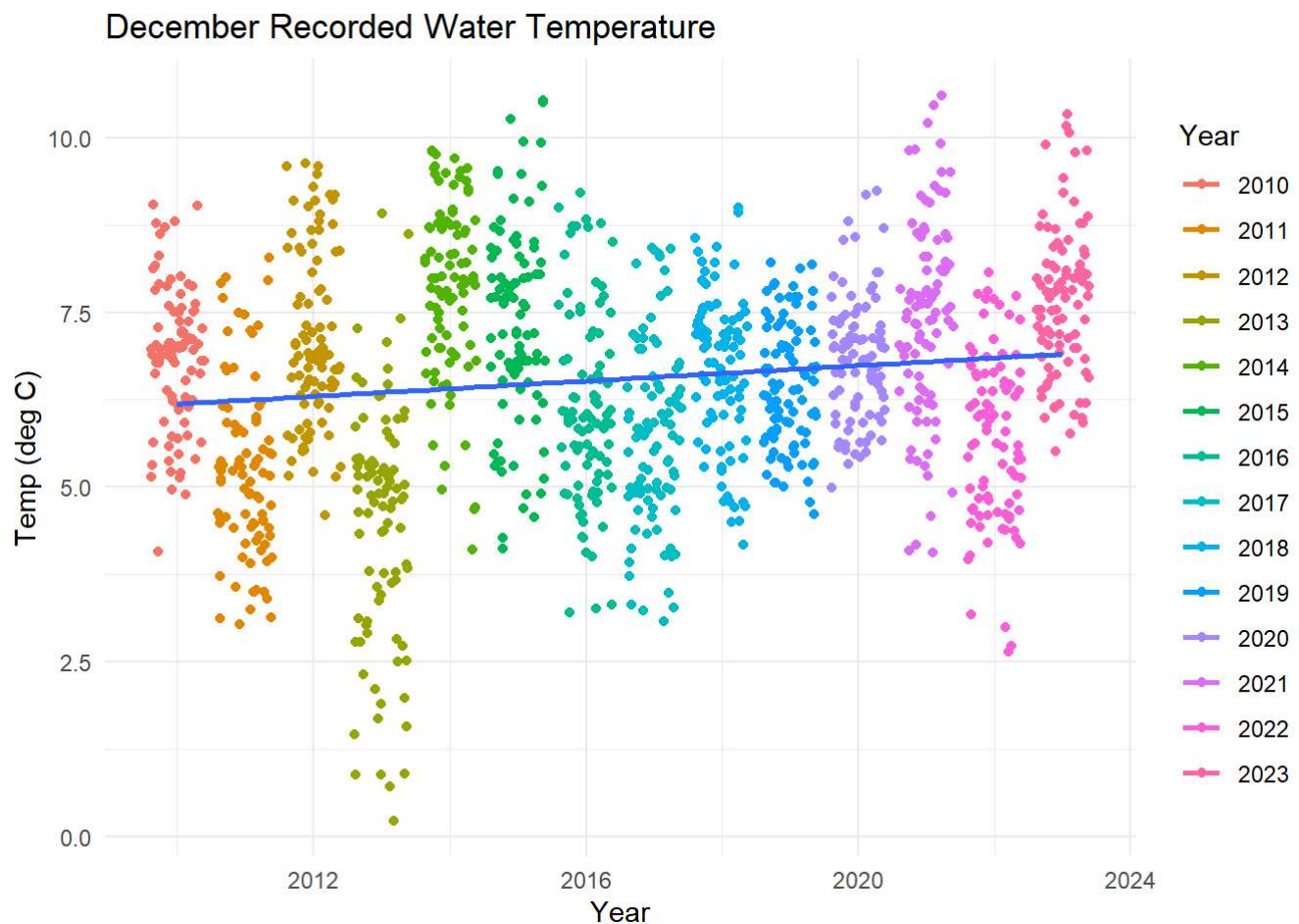
Null deviance: 3137.6 on 1298 degrees of freedom
 Residual deviance: 2909.3 on 1295 degrees of freedom
 AIC: 4743.8

Number of Fisher Scoring iterations: 2

```
dec_plot <- ggplot(dec_data, aes(x = year,
                                y = temp_C,
                                color = as.factor(year))) +

  theme_minimal() +
  geom_jitter() +
  geom_smooth(aes(group = 1),
              method = "glm",
              method.args = list(family = "gaussian"),
              se = F) +
  labs(title = "December Recorded Water Temperature",
       x = "Year",
       y = "Temp (deg C)",
       color = "Year")

dec_plot
```



PART III: MANAGEMENT RECOMMENDATIONS: SEASONAL WATER TEMPERATURE CHANGES

How large are the seasonal temperature differences, and do they change over time?

```
#|message: false

#get data for August and December
summary_temp_data <- temp_data_final |>
  clean_names() |>
  filter(month %in% c("Aug", "Dec")) |>
  select(site_no, date, temp_c, month, year)

#calculate mean water temp per month per year
mean_temps <- summary_temp_data |>
  group_by(year, month) |>
  summarise(
    mean_temp = mean(temp_c),
  )
```

`summarise()` has grouped output by 'year'. You can override using the `.groups` argument.

```
mean_temps
```

```
# A tibble: 28 × 3
# Groups:   year [14]
   year month mean_temp
  <dbl> <chr>   <dbl>
1  2010 Aug      19.4
2  2010 Dec       6.86
3  2011 Aug      19.0
4  2011 Dec       5.26
5  2012 Aug      19.6
6  2012 Dec       7.13
7  2013 Aug      19.8
8  2013 Dec       4.50
9  2014 Aug      20.4
10 2014 Dec       7.87
# i 18 more rows
```

```
diff_temps <- mean_temps |>
  pivot_wider(names_from = month, values_from = mean_temp) |>
  clean_names() |>
  mutate(temp_diff = aug - dec,
         temp_diff = round(temp_diff, 2))
diff_temps
```

```
# A tibble: 14 × 4
# Groups:   year [14]
   year  aug  dec temp_diff
  <dbl> <dbl> <dbl>   <dbl>
1  2010  19.4  6.86    12.5
2  2011  19.0  5.26    13.7
3  2012  19.6  7.13    12.5
4  2013  19.8  4.50    15.3
5  2014  20.4  7.87    12.6
6  2015  21.3  7.35    14.0
7  2016  21.1  6.14    14.9
8  2017  19.9  5.73    14.2
9  2018  20.7  6.68    14.0
10 2019  20.6  6.46    14.2
11 2020  20.5  6.86    13.6
12 2021  21.5  7.43    14.0
13 2022  21.1  5.69    15.4
14 2023  21.4  7.65    13.8
```

```
min(diff_temps$temp_diff) # equals 12.47
```

```
[1] 12.47
```

```
mean(diff_temps$temp_diff) # equals 13.90
```

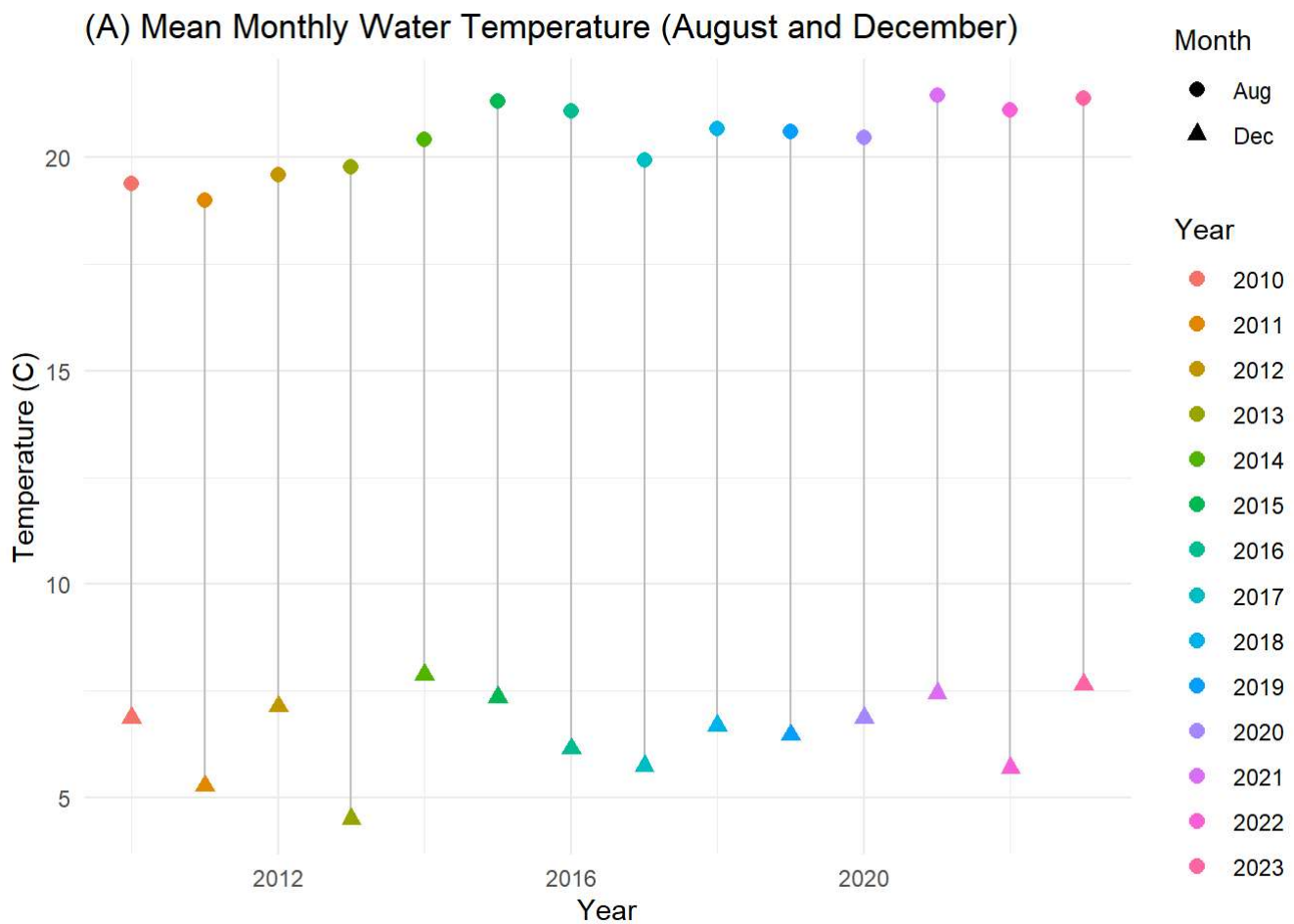
```
[1] 13.90429
```

```
max(diff_temps$temp_diff) # equals 15.42
```

```
[1] 15.42
```

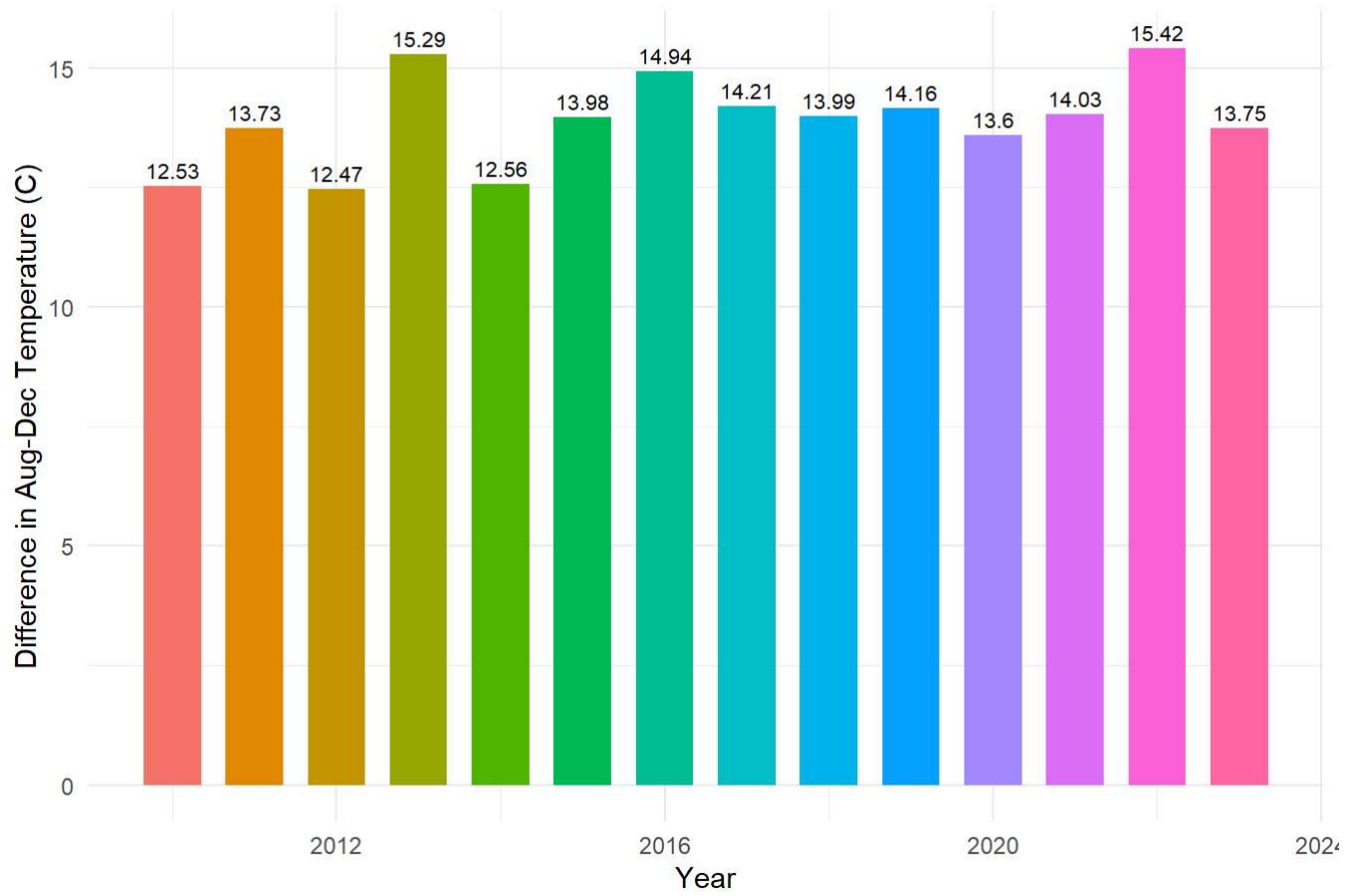
```
# Plot seasonal temperature means
```

```
mean_temps_plot <- ggplot(mean_temps, aes(x = year,  
                                           y = mean_temp,  
                                           color = as.factor(year))) +  
  stat_summary(fun.min = min, fun.max = max, geom = "linerange", color = "grey") +  
  geom_point(aes(shape = as.factor(month)), size = 2.5) +  
  labs(x = "Year",  
       y = "Temperature (C)",  
       title = "(A) Mean Monthly Water Temperature (August and December)",  
       color = "Year",  
       shape = "Month") +  
  theme_minimal()  
mean_temps_plot
```



```
# plot annual difference in seasonal temperatures
diff_plot <- ggplot(diff_temps, aes(x = year, y = temp_diff)) +
  geom_bar(stat = "identity", width = 0.7, aes(fill = as.factor(year))) +
  geom_text(aes(label = round(temp_diff, 2), vjust = -0.5), size = 2.8) +
  labs(x = "Year",
       y = "Difference in Aug-Dec Temperature (C)",
       title = "(B) Seasonal Difference in Water Temperature") +
  guides(fill = "none") +
  theme_minimal()
diff_plot
```

(B) Seasonal Difference in Water Temperature



```
library(patchwork)
mean_temps_plot + diff_plot
```

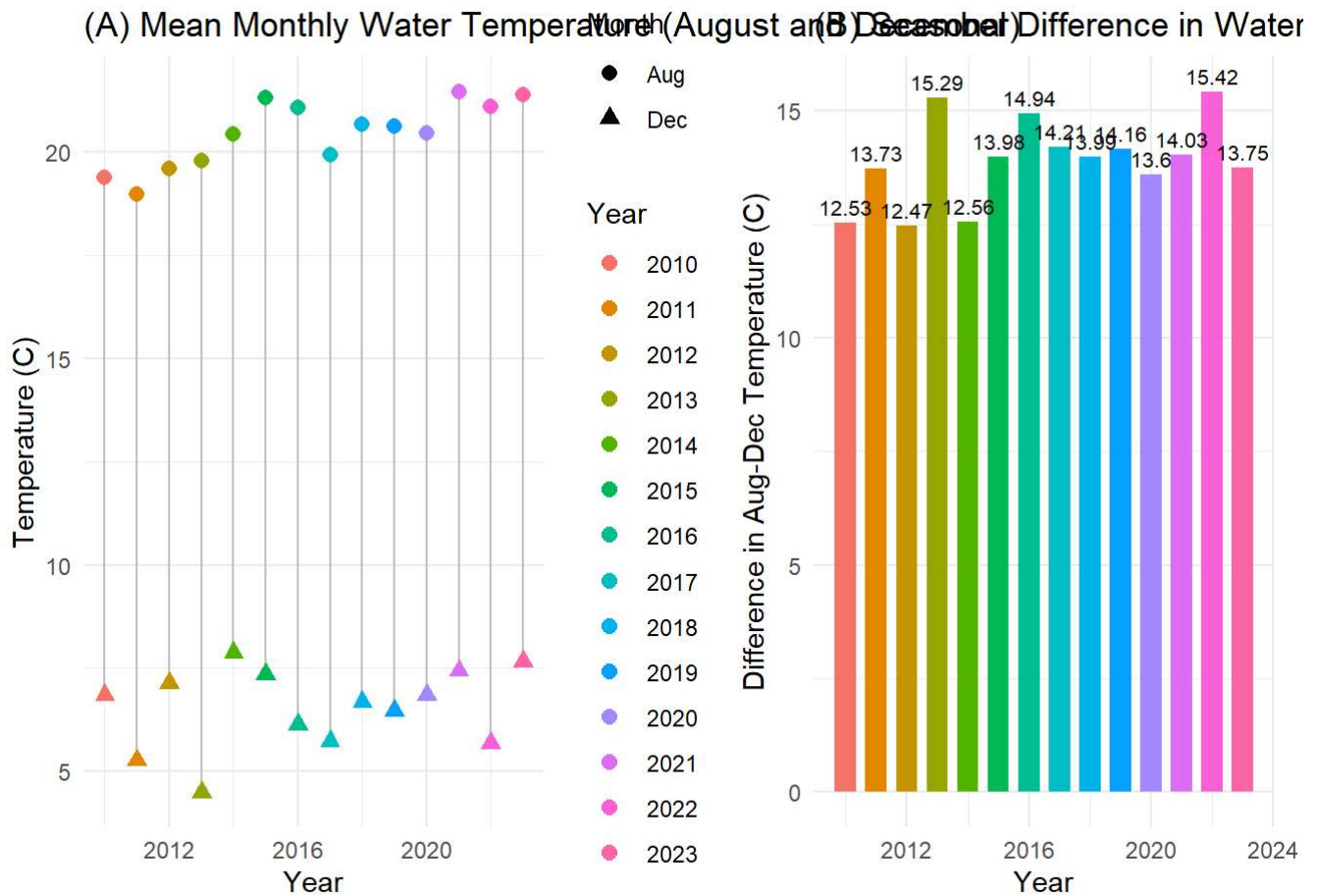


Figure A above shows the difference in mean water temperature between August and December annually from 2010 through 2023. The seasonal temperature difference between August and December ranges from 12.47 deg C to 15.42 deg C, with a mean of 13.90 deg C difference. For conservation planning purposes, it is reasonable to estimate that the water temperature will decrease approximately 13-14 degrees between the summer and the winter this season. In addition, Figure B plots the water temperature difference over time. There are no clear outliers, and the distribution is relatively uniform. Therefore, it is expected that near-term seasonal water temperature decreases will follow a similar pattern. Even though the analysis of August and December data shows that the temperatures are increasing each year, they appear to be increasing at a consistent rate so that the seasonal change stays relatively constant.